



Manipulating high-intensity interval training: Effects on $\dot{V}O_{2\max}$, the lactate threshold and 3000 m running performance in moderately trained males

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KEYWORDS

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Summary The aim of this study was to compare the effects of two high-intensity interval training (HIT) programmes on maximal oxygen uptake ($\dot{V}O_{2\max}$), the lactate threshold (LT) and 3000 m running performance in moderately trained male runners. $\dot{V}O_{2\max}$, the running speed associated with $\dot{V}O_{2\max}$ ($v_{\dot{V}O_{2\max}}$), the time for which $v_{\dot{V}O_{2\max}}$ can be maintained ($T_{\max}$), the running speed at LT (v_{LT}) and 3000 m running time (3000mTT) were determined before and following three different training programmes performed for 10 weeks. Following the pre-test, 17 moderately trained male runners ($\dot{V}O_{2\max} = 51.6 \pm 2.7 \text{ ml kg}^{-1} \text{ min}^{-1}$) were divided into training groups based on their 3000mTT (Group 1, G_1 , $N=6$, $8 \times 60\%$ of $T_{\max}$ at $v_{\dot{V}O_{2\max}}$, 1:1 work:recovery ratio; Group 2, G_2 , $N=6$, $12 \times 30\text{s}$ at $130\% v_{\dot{V}O_{2\max}}$, 4.5 min recovery; control group, G_{CON} , $N=5$, 60 min at $75\% v_{\dot{V}O_{2\max}}$). G_1 and G_2 performed two HIT sessions and two 60 min recovery run sessions ($75\% v_{\dot{V}O_{2\max}}$) each week. Control subjects performed four 60 min recovery run sessions ($75\% v_{\dot{V}O_{2\max}}$) each week. In G_1 , significant improvements ($p < 0.05$) following HIT were found in $\dot{V}O_{2\max}$ (+9.1%), $v_{\dot{V}O_{2\max}}$ (+6.4%), $T_{\max}$ (5%), v_{LT} (+11.7%) and 3000mTT (−7.3%). In G_2 , significant improvements ($p < 0.05$) following HIT were found in $\dot{V}O_{2\max}$ (+6.2%), $v_{\dot{V}O_{2\max}}$ (+7.8%), $T_{\max}$ (+32%) and 3000mTT (−3.4%), but not in v_{LT} (+4.7%; $p = 0.07$). No significant changes in these variables were found in G_{CON} . The present study has shown that 3000 m running performance, $\dot{V}O_{2\max}$, $v_{\dot{V}O_{2\max}}$, $T_{\max}$ and v_{LT} can be significantly enhanced using different HIT programmes in moderately trained runners, but that changes in performance and physiological variables may be more profound using prolonged HIT at intensities of $v_{\dot{V}O_{2\max}}$ with interval durations of $60\% T_{\max}$.

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Introduction

Various types of high-intensity interval training (HIT) programmes have been shown to improve endurance performance and associated physiological variables.^{1,2} The magnitude of the training response appears to depend upon the duration, intensity and frequency of the interval bouts, as well as the type (active or passive) and duration of the recovery period between the exercise bouts.³ However, little is known concerning the magnitude of the change in performance and associated physiological variables that may be obtained following different types of prescribed training intensities and durations.^{1,2,4}

It has been suggested that the $v_{\dot{V}O_{2\max}}$, defined as the minimum speed needed to reach $\dot{V}O_{2\max}$,⁵ might represent an optimal training stimulus when the goal is to run for as long as possible at $\dot{V}O_{2\max}$.⁶ Hill and Rowell⁷ contend that training at $v_{\dot{V}O_{2\max}}$ is important because $v_{\dot{V}O_{2\max}}$ is the lowest speed that will elicit $\dot{V}O_{2\max}$ and it may be optimal to train at $\dot{V}O_{2\max}$ in order to improve it. The $v_{\dot{V}O_{2\max}}$ has been shown to be a good discriminator of the aerobic performance potential of middle and long distance runners,^{8–10} and the $v_{\dot{V}O_{2\max}}$ may be more closely associated with race pace in middle distance events compared to the slower velocities of longer distance races.¹¹ A concept that is closely related to $v_{\dot{V}O_{2\max}}$ is the time for which exercise at $v_{\dot{V}O_{2\max}}$ can be sustained ($T_{\max}$).^{4,11} Previous studies suggest that HIT performed somewhere between 50 and 60% of $T_{\max}$ may be an optimal HIT bout duration for improving $\dot{V}O_{2\max}$ and endurance performance.^{4,11,12}

Although repeated supramaximal HIT has not been traditionally used in aerobic endurance training programmes, recent work has shown that this type of training can improve endurance performance in well-trained cyclists.^{2,13} Both Stepto et al.¹³ and Laursen et al.² showed that repeated supramaximal HIT (30 s at 170% of peak power output, 4.5 min recovery) provided significant gains (+4%) in 40-km cycling time trial performance. Even more recently, Burgomaster et al.¹⁴ demonstrated in recreationally active subjects that six sprint interval training sessions (30 s ‘‘all-out’’ Wingate tests, 4 min recovery) performed over 2 weeks increased citrate synthase activity by 38% and doubled endurance capacity during cycling exercise at $\sim 80\% \dot{V}O_{2\text{peak}}$. Thus, while repeated supramaximal HIT appears to be an effective means of enhancing cycling performance, the effect of repeated supramaximal HIT has, to our knowledge, not been

examined longitudinally in runners. Moreover, a comparison of supramaximal HIT versus a HIT programme prescribed at $v_{\dot{V}O_{2\max}}$ intensity on performance and physiological variables are lacking in runners. Indeed, these two types of described interval training programmes are vastly different, and while both may improve endurance running performance and physiological variables, a comparison of the magnitude of change received from such programmes would be of interest to sports scientists and coaches alike.

The purpose of this study therefore was to compare the influence of a supramaximal HIT programme with a HIT programme using intervals at $v_{\dot{V}O_{2\max}}$ intensity for 60% of $T_{\max}$ on 3000 m running performance, $\dot{V}O_{2\max}$ and the lactate threshold in moderately trained male runners. We hypothesized that both supramaximal HIT and HIT prescribed using intervals at $v_{\dot{V}O_{2\max}}$ intensity for 60% of $T_{\max}$ would result in significantly improved 3000 m time trial performance, $\dot{V}O_{2\max}$ and the lactate threshold compared with low intensity controls, but that performance and physiological gains from HIT at $v_{\dot{V}O_{2\max}}$ would be more pronounced.

Methods

Subjects

Seventeen moderately trained male runners that had 2–3 years of run training experience volunteered to participate in this study. The subjects had not performed any HIT for a minimum of 3 months before this investigation, and their average training distance during the study was $38 \pm 4 \text{ km week}^{-1}$, which was similar to their training distance before the study. Subjects had the following characteristics: age = 19 ± 2 years; height = $172 \pm 4 \text{ cm}$; mass = $73 \pm 3 \text{ kg}$. The participants provided written informed consent to the experimental procedures after the possible benefits and risks of participation were explained to them. The study protocol was approved by the human research ethics committee of the Tehran Teacher Training University.

Experimental protocol and procedures

The participants were instructed to arrive at the laboratory in a rested and hydrated state and to avoid strenuous exercise in the 48 h period preceding a test session. Subjects were also asked to complete food diaries on the two days before baseline testing and to replicate this diet before the

Table 1 Ten-week training protocol for the two high-intensity interval training groups (G_1 , G_2) and the control group (G_{CON})

Group	Sessions/week	Bouts/session	Intensity	Work duration	Rest duration
G_1	2	8	$\dot{V}O_{2\max}$	60% $T_{\max}$	60% $T_{\max}$
($N=6$)	2	1	75% $\dot{V}O_{2\max}$	60 min	NA
G_2	2	12	130% $\dot{V}O_{2\max}$	30 s	4.5 min
($N=6$)	2	1	75% $\dot{V}O_{2\max}$	60 min	NA
G_{CON} ($N=5$)	4	1	75% $\dot{V}O_{2\max}$	60 min	NA

$\dot{V}O_{2\max}$: minimal speed to elicit $\dot{V}O_{2\max}$; $T_{\max}$ = time to exhaustion at $\dot{V}O_{2\max}$. All training groups performed four training sessions per week.

post-training test session. Two weeks prior to the start of the testing and training programme, subjects reported to the laboratory for four treadmill running familiarisation trials, consisting of 60 min of running at a self-selected pace on the treadmill. Subjects were also shown the metabolic cart and were allowed to wear the breathing apparatus and heart rate monitor. Subjects completed their performance tests on three separate days. The first day of testing was a progressive exercise test used to determine $\dot{V}O_{2\max}$, $\dot{V}O_{2\max}$, and the speed at lactate threshold (v_{LT}). The second test performed was a $T_{\max}$ test used to determine the time to exhaustion at $\dot{V}O_{2\max}$. Subjects also completed a 3000 m running time trial on a 200 m indoor synthetic running track. All tests were separated by ≥ 48 h and were completed within a 1-week period. Following the first week of testing, subjects were matched into three separate training groups based on their 3000 m run performance time as follows: Group 1, G_1 ($N=6$); Group 2, G_2 ($N=6$); control group, G_{CON} ($N=5$). All subjects undertook a 10-week treadmill training programme. The HIT groups (G_1 and G_2) performed two HIT sessions and two 60 min recovery run sessions (75% $\dot{V}O_{2\max}$) each week. Control subjects performed four 60 min recovery run sessions (75% $\dot{V}O_{2\max}$) per week (Table 1). Following the 10-week treadmill training programmes, subjects repeated the same laboratory and field tests as the pre-training tests in the same order and under the same conditions. All laboratory testing and training sessions were carried out on a running treadmill (Tunturi T 807, Finland) at 19–23 °C in the Physiology Laboratory at the Tehran Teacher Training University. During the training and testing period, all subjects reported that they did no other training.

Determination of $\dot{V}O_{2\max}$, $\dot{V}O_{2\max}$ and v_{LT}

Subjects performed a progressive treadmill exercise test whereby the initial speed was set at

8 km h⁻¹, and speed increased by 1 km h⁻¹ every 3 min until exhaustion. All stages were followed by 30 s of rest¹⁵ during which a fingertip capillary blood sample was collected to determine blood lactate concentration (Lactate Pro, Arkray, Japan). Cardiorespiratory-metabolic variables were measured using a telemetric system (Meta-max 3B, United Kingdom) that was calibrated before and verified after each test according to the manufacturer's instructions. During the progressive exercise test, each subject was encouraged to give a maximal effort.

$\dot{V}O_{2\max}$ was recorded as the highest $\dot{V}O_2$ value reached during the incremental test. $\dot{V}O_{2\max}$ was confirmed when three or more of the following criteria were met: (1) a plateau in $\dot{V}O_2$ despite an increase in running speed; (2) a respiratory exchange ratio (RER) higher than 1.20; (3) a heart rate within 10 bpm of its predicted maximum; (4) a lactate concentration higher than 10 mmol l⁻¹; and/or (5) visible subject exhaustion.¹⁶ $\dot{V}O_{2\max}$ was defined as the minimal speed at which the athlete was running when $\dot{V}O_{2\max}$ occurred (within 2 ml kg⁻¹ min⁻¹ of $\dot{V}O_{2\max}$ ¹²), as long as this speed was sustained for at least 1 min.¹⁷ If an athlete achieved $\dot{V}O_{2\max}$ during a stage that was not sustained for 1 min, the speed of the previous stage was recorded as $\dot{V}O_{2\max}$.¹⁷ Lactate threshold (v_{LT}) was identified as the speed when the first significant elevation of blood lactate above resting levels occurred.^{18,19}

Determination of $T_{\max}$

After a warm-up consisting of a 5 min run at 50% $\dot{V}O_{2\max}$, 5 min stretching and another 5 min run at 60% $\dot{V}O_{2\max}$, the treadmill speed was increased to $\dot{V}O_{2\max}$ and the stopwatch was started. Subjects were verbally encouraged to run for as long as possible. The test was terminated by the subject and the time running at $\dot{V}O_{2\max}$ ($T_{\max}$) was recorded.^{4,11} The post-HIT $T_{\max}$ test was run at the

same speed as the pre-HIT $T_{\max}$ test under the same conditions.

Determination of 3000 m time trial performance

The 3000 m time trial was run on an indoor 200 m track. As it was not possible to run all 17 subjects at the same time, subjects were divided into three groups according to their availability for testing and performance potential. The strategy of running athletes of similar ability at the same time was considered to be the best way of optimising each subject's performance.²⁰ The subjects were given a warm-up which included a 5 min run at 50% $\dot{V}O_{2\max}$, 5 min stretching and another 5 min of running at 50% $\dot{V}O_{2\max}$. Subjects were informed of their lap times throughout the time trial.

High-intensity interval training programmes

Athletes were evenly matched into groups based on their 3000 m running performance (Table 1).² The two HIT groups performed HIT twice per week for 10 weeks. At each HIT session, runners in G_1 completed five to eight intervals at $\dot{V}O_{2\max}$ for a duration equal to 60% $T_{\max}$, with a 1:1 work:recovery ratio. The HIT sessions for G_1 included high-intensity running bouts at 15.7 (0.7) km h⁻¹ for 3.5 (0.7) min followed by low-intensity recovery runs at 7.8 (0.3) km h⁻¹ for 3.5 (0.7) min. G_2 completed seven to twelve 30 s bouts at 130% $\dot{V}O_{2\max}$ separated by 4.5 min of recovery. This equated to a supramaximal run at 19.9 (0.6) km h⁻¹ for 30 s followed by a recovery run at 7.8 (0.3) km h⁻¹ for 4.5 min (Table 1). Each HIT session lasted approximately 60 min for both groups. Although the training intensity for both groups did not change throughout the training programme, the number of fully completed HIT bouts was increased in G_1 from 4.8 ± 1.0 to 7.5 ± 0.7 repetitions and in G_2 from 7.5 ± 1.2 to 9.0 ± 2.6 repetitions. All HIT sessions were separated by at least 2 days. Subjects in the HIT groups also completed two recovery runs per week, which consisted of a self-selected warm-up, a 60 min treadmill run at 75% $\dot{V}O_{2\max}$ (11.5 ± 0.5 km h⁻¹) and a cool down. Control subjects performed four 60 min low-intensity (75% $\dot{V}O_{2\max}$; 11.5 ± 0.5 km h⁻¹) running sessions per week (Table 1). During the study, subjects did no other training, and the total weekly running distance on the last week of the training programme was similar in all three groups ($G_1 = 43.6$ (2.4) km week⁻¹; $G_2 = 36.6$

(2.6) km week⁻¹; $G_{\text{CON}} = 46$ (2.2) km week⁻¹). The primary investigator was present for and controlled all training and testing sessions.

Data analysis

The equality of variance was verified prior to performing parametric analyses. A paired Student's t -test was run separately in each specific training group in order to determine whether each specific HIT programme had any effect on the dependent measures. Differences between groups were analysed using a 3×2 (group $\times$ time) repeated measures analysis of variance (ANOVA) followed by Tukey's post hoc test when a significant F -ratio was observed. Pearson product moment (r) examined relationships between variables, while stepwise linear regression was used to examine the physiological variables that were best related to 3000 m running performance. Significance was set at $p \leq 0.05$ for all analyses.

Results

The 3000 m time trial performance

The 3000 m time trial performance time was significantly decreased post-HIT in G_1 ($p < 0.001$) and G_2 ($p < 0.01$) but not in G_{CON} (Fig. 1). The 3000 m time trial performance improved by 50 s in G_1 (-7.3%), 23 s in G_2 (-3.4%) and 1 s in G_{CON} (-0.1%). When groups were compared, the improvement in 3000 m time trial performance in G_1 was significantly greater than the improvement in G_{CON} ($p < 0.05$). Although a trend was shown for a greater improvement in 3000 m time trial performance in G_2 compared to G_{CON} ($p = 0.27$), this was not statistically significant (Fig. 1).

$\dot{V}O_{2\max}$, $\dot{V}O_{2\max}$ and $T_{\max}$

$\dot{V}O_{2\max}$ significantly increased in G_1 and G_2 , but not in G_{CON} (Table 2). As well, the increase in $\dot{V}O_{2\max}$ was significantly greater in G_1 compared with the increase in G_{CON} ($p < 0.05$). Although $\dot{V}O_{2\max}$ in G_2 was significantly increased by 6.2%, this change was not significantly different compared with the change in $\dot{V}O_{2\max}$ in G_{CON} ($p = 0.18$). $\dot{V}O_{2\max}$ and $T_{\max}$ were both significantly increased in G_1 and G_2 , but not in G_{CON} (Table 2). As well, the change in $\dot{V}O_{2\max}$ and $T_{\max}$ in both training groups was significantly greater compared with the change in $\dot{V}O_{2\max}$ and $T_{\max}$ in G_{CON} (Table 2).

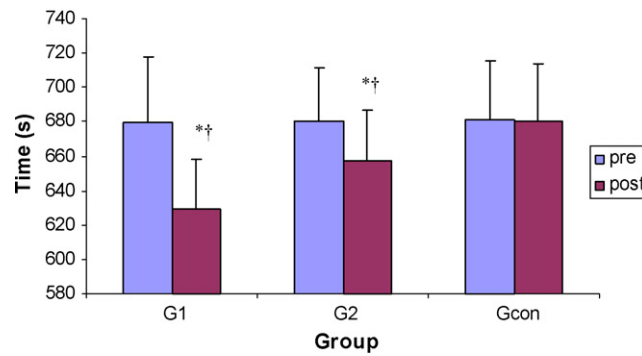


Figure 1 Pre-training vs. post-training 3000 m running time in Group 1 (G_1), Group 2 (G_2) and the control group (G_{CON}). *Significantly lower than pre-training ($p < 0.05$). †Significantly different from control group ($p < 0.05$).

Table 2 Pre-training vs. post-training values for maximal oxygen uptake ($\dot{V}O_{2\max}$), running speed at $\dot{V}O_{2\max}$ ($v_{\dot{V}O_{2\max}}$), the time for which this running speed can be maintained ($T_{\max}$) and the running speed at the lactate threshold (v_{LT}) in the different training groups

	Group		
	G_1	G_2	G_{con}
$\dot{V}O_{2\max}$ (ml kg ⁻¹ min ⁻¹)			
Pre	51.3 (2.4)	51.7 (3.4)	51.8 (2.8)
Post	56.0 (1.4)*,†	54.9 (2.1)*	52.9 (1.5)
%Δ	+9.1	+6.2	+2.1
$v_{\dot{V}O_{2\max}}$ (km h ⁻¹)			
Pre	15.6 (0.7)	15.4 (0.5)	15.3 (0.4)
Post	16.6 (0.6)*,†	16.6 (0.6)*,†	15.5 (0.3)
%Δ	+6.4	+7.8	+1.3
$T_{\max}$ (s)			
Pre	330 (21)	308 (17)	285 (18)
Post	447 (19)*,†	407 (16)*,†	295 (18)
%Δ	+35	+32	+3.5
v_{LT} (km h ⁻¹)			
Pre	10.2 (0.7)	10.5 (0.5)	10.6 (0.4)
Post	11.4 (0.6)*	11.0 (0.6)	10.8 (0.3)
%Δ	+11.7	+4.7	+1.9

Data are means (±S.D.).

* Significantly greater than pre-training value ($p < 0.05$).

† Significantly different change compared with control group ($p < 0.05$).

Lactate threshold

There were significant increases in v_{LT} in G_1 (Table 2). While G_2 showed a 0.5 km h^{-1} improvement in v_{LT} , this change was not statistically significant ($p = 0.07$). Moreover, the improvement in v_{LT} in G_1 and G_2 was not significantly different than the improvement in v_{LT} in G_{CON} (Table 2).

Correlations

Before training, 3000 m running performance time was significantly related to $\dot{V}O_{2\max}$ ($r = -0.77$),

$v_{\dot{V}O_{2\max}}$ ($r = -0.76$) and v_{LT} ($r = -0.63$), but not to $T_{\max}$ ($r = -0.31$; $N = 17$). After training, these correlations tended to improve, and 3000 m running performance time was significantly related to $\dot{V}O_{2\max}$ ($r = -0.83$), $v_{\dot{V}O_{2\max}}$ ($r = -0.81$) and v_{LT} ($r = -0.67$), but not to $T_{\max}$ ($r = -0.37$). The change in 3000 m running time was significantly related to the change in $\dot{V}O_{2\max}$ ($r = 0.76$), v_{LT} ($r = 0.64$) and $T_{\max}$ ($r = 0.48$), but not to $v_{\dot{V}O_{2\max}}$ ($r = 0.42$; $p = 0.09$). Stepwise linear regression analysis revealed that the only variable related to the change in 3000 m running performance time was the change in $\dot{V}O_{2\max}$ ($R = 0.76$; $p < 0.001$; $N = 17$).

Discussion

The important findings of the present study were that: (1) HIT prescribed using $v_{\dot{V}O_{2\max}}$ as the interval intensity and 60% of $T_{\max}$ as the interval duration improved 3000 m running performance time (−7.3%), concomitant with increases in $\dot{V}O_{2\max}$ (+9.1%), $v_{\dot{V}O_{2\max}}$ (+6.4%), $T_{\max}$ (+35%) and v_{LT} (+11.7%); (2) repeated 30 s maximal sprint interval training improved 3000 m running performance (−3.4%) with simultaneous increases in $\dot{V}O_{2\max}$ (+6.2%), $v_{\dot{V}O_{2\max}}$ (+7.8%) and $T_{\max}$ (+32%), but not v_{LT} (+4.7%; $p=0.07$); (3) the improvements in performance and related physiological variables tended to be greater (in absolute, but not statistical terms) using more prolonged HIT at $v_{\dot{V}O_{2\max}}$ when compared with supramaximal sprint interval training. These general findings will be elaborated on.

The first finding of the present study was that HIT prescribed using $v_{\dot{V}O_{2\max}}$ as the interval intensity, and 60% of $T_{\max}$ as the interval duration, improved 3000 m running performance time by 50 s (−7.3%; $p<0.05$) in moderately trained male runners (Fig. 1). These performance improvements are greater than those reported by Smith et al.^{4,11} who reported a 17 s (−2.7%) improvement in 3000 m time trial performance in well-trained runners using HIT prescribed at $v_{\dot{V}O_{2\max}}$ with interval durations of 60–70% of $T_{\max}$. The greater improvement in 3000 m time trial performance shown in the present study may be due to differences in the training level of the subjects and/or the differences in the duration of the HIT programmes (i.e., 4 weeks versus 10 weeks in the present study). Regardless, the present findings demonstrate that HIT prescribed using $v_{\dot{V}O_{2\max}}$ as the interval intensity with interval durations at 60% of $T_{\max}$ are an effective means of enhancing 3000 m running performance time in moderately trained male runners.

Another important finding from the present study was that the repeated supramaximal HIT programme of G_2 lead to a 23 s (−3.4%; $p<0.05$) improvement in 3000 m running performance time (Fig. 1). While supramaximal HIT has been shown to improve 40-km cycle time trial performance in well-trained cyclists,^{2,13} and cycle time to exhaustion at 80% $\dot{V}O_{2\max}$ in untrained subjects,¹⁴ this is to our knowledge the first study to show improvements in endurance running performance following supramaximal HIT in moderately trained male runners. Although the 23 s improvement in performance following supramaximal HIT in G_2 is impressive, this was not as great in absolute terms as the 50 s improvement found for G_1 (Fig. 1).

Although not invasively examined in the present study, some discussion of the potential mechanisms that may have been responsible for the improvements in 3000 m running performance and associated physiological variables found in G_1 and G_2 is warranted. It has been suggested that the relative contribution of aerobic and anaerobic metabolism to a 10 min 3000 m run is approximately 85 and 15%, respectively.²⁰ Therefore, improvement of 3000 m running performance may occur through increases in the metabolic capacity of either the aerobic or anaerobic energy systems, or both.^{21,22} The present study found that HIT improved the $\dot{V}O_{2\max}$ of the runners in G_1 (+9.1%) and G_2 (+6.2%) (Table 2). While the increase in $\dot{V}O_{2\max}$ in G_1 was significantly greater than the increase found in G_{CON} (+2.1%; $p<0.05$), the 6.2% enhancement in $\dot{V}O_{2\max}$ found in G_2 was not ($p=0.18$). Nevertheless, these findings are suggestive that some of the improvement found in 3000 m running performance time was likely related to an increase in aerobic capacity, which is further supported through the finding that the change in 3000 m running time was significantly related to the change in $\dot{V}O_{2\max}$ ($R=0.76$; $p<0.001$; $N=17$). It is also possible that the greater improvement in 3000 m running performance time in G_1 versus G_2 was aided through the slightly higher training volume in G_1 (43.6 km week^{−1}) compared with G_2 (36.6 km week^{−1}).

Several studies have shown an increase in $\dot{V}O_{2\max}$ (+5 to 15%) following various HIT programmes.^{2,11,23–28} Improvements in $\dot{V}O_{2\max}$ may occur through increases in both oxygen delivery (i.e., increases in stroke volume) and/or oxygen utilisation by active muscles (i.e., increases in capillarisation/mitochondrial density).³ The magnitude of the increase in $\dot{V}O_{2\max}$ appears to depend upon the training stimulus (intensity, duration and frequency of the training programme), the starting level of fitness and one's genetic response to the training stimulus.¹ The finding of a greater absolute (but non-significant) increase in $\dot{V}O_{2\max}$ in G_1 compared with G_2 (Table 2) is in agreement with the findings of Laursen et al.² who noted a greater improvement in $\dot{V}O_{2\max}$ with HIT prescribed at 60% $T_{\max}$ (+5.4 to +8.1%) compared with supramaximal HIT (+3%) in well-trained cyclists. Similarly, Franch et al.²⁴ reported in recreational runners a 6% increase in $\dot{V}O_{2\max}$ after long interval training (6× 4 min at 16.5 km h^{−1}, 2 min recovery) but only a 3% improvement with short interval training (30× 15 s at 20.5 km h^{−1}, 15 s recovery). Thus, a HIT programme that requires substantial oxygen delivery (i.e., G_1 ¹²) may induce adaptations to the oxygen delivery system (likely through increases in stroke volume/cardiac output),¹

that likely leads to improvements in endurance performance (i.e., Fig. 1). Other adaptations are also likely responsible for the improvement in endurance performance including the ability to utilise oxygen efficiently, neuromuscular system adaptations, and improvements in running economy.

The finding of improvements in both 3000 m running performance (Fig. 1) and $\dot{V}O_{2\max}$ (Table 2) following repeated supramaximal HIT (i.e., G₂) is similar to the findings of Laursen et al.² who reported a 4.4% improvement in 40-km cycle time trial performance coupled with a 3% increase in $\dot{V}O_{2\max}$ in well-trained cyclists. In direct contrast to these findings, Burgomaster et al.¹⁴ recently showed in untrained subjects that six sessions of sprint interval training (all out 30 s Wingate sprints) increased citrate synthase activity by 38% and doubled cycle time to exhaustion at 80% $\dot{V}O_{2\max}$. Interestingly, no change occurred in $\dot{V}O_{2\max}$. However, the supramaximal HIT programme of the present study and that of Laursen et al.² were performed for a greater total duration compared with the study of Burgomaster et al.¹⁴ Moreover, subjects in G₂ of the present study performed two additional 60 min 75% $\dot{V}O_{2\max}$ training runs each week (Table 1), and the subjects in the study of Laursen et al.² performed additional submaximal endurance training. It may be, therefore, that a greater training duration and volume is required before central or peripheral adaptations related to $\dot{V}O_{2\max}$ occur. Nevertheless, the present study is in agreement with that of others^{2,14,26,29,30} with respect to the fact that repeated supramaximal HIT can improve endurance performance. The precise mechanisms involved, however, remain undetermined. Potential adaptations that may contribute to the improvement in endurance performance following HIT include the ability of skeletal muscle to buffer hydrogen ions³¹ as well as increases in Na⁺/K⁺ pump capacity,³² glycolytic flux,²⁶ anaerobic capacity^{22,33} and/or motor unit activation.²⁸

Other notable adaptations following the HIT programmes of the present study included enhancements in $\dot{V}O_{2\max}$, $T_{\max}$ and v_{LT} (Table 2). The $v_{\dot{V}O_{2\max}}$, or running speed associated with $\dot{V}O_{2\max}$, is a useful variable that combines $\dot{V}O_{2\max}$ and economy of motion into a single factor that can segregate aerobic and performance differences between runners.⁹ In the present study, $v_{\dot{V}O_{2\max}}$ increased significantly in G₁ (+6.4%) and G₂ (+7.8%) compared with only a 1.3% increase in G_{CON} (Table 2). Such results are in accordance with previous studies reporting an improvement in $v_{\dot{V}O_{2\max}}$ (+3 to 10%) after HIT in subjects of differing aerobic training levels.^{11,17,23,24}

It has been suggested that neural adaptations following sprint training are causal for improvements in $v_{\dot{V}O_{2\max}}$ and running economy.²⁸ Although running economy was not measured in the present study, it is possible that the greater improvements in $v_{\dot{V}O_{2\max}}$ in G₂ may have been due to improvements in running economy.^{17,28}

Blood lactate levels are linked to the muscle respiratory capacity,²³ and the ability to produce less lactate at a given running speed is a determinant of prolonged running performance.³⁴ A right shift in the lactate threshold so that higher running speeds are achieved at equivalent blood lactate levels is a well-known consequence of endurance training. It has been suggested that the submaximal blood lactate response to exercise is associated with peripheral factors including muscle fibre type, capillary density and the number of mitochondria.³ In the present study, v_{LT} was increased in G₁ by 11.7% (Table 2). The increased mitochondrial enzyme content, which is associated with an increase in the rate of lipid utilisation and consequently a decrease in rate of glycogen depletion,³ is one possible explanation for the right shift in the lactate threshold speed found after the HIT programmes employed in the present study.¹ An increased capillary density after endurance training could also increase the exchange area and decrease the distance between the site of lactate production and the capillary wall, which could improve the lactate exchange ability.³⁴ Data from the present study indicate that the $v_{\dot{V}O_{2\max}}$ intervals had a beneficial effect on $\dot{V}O_{2\max}$ and the lactate threshold (Table 2). Although, v_{LT} was increased in G₂ by 4.7%, this increase was not significantly different from the pre-test variables ($p=0.07$), likely due to the small sample size. Indeed, the low sample size used in the groups of the present study can lead to Type II statistical errors.

In conclusion, the present study has extended the findings of previous investigations^{4,11,17} in showing that 3000 m running performance, $\dot{V}O_{2\max}$, $v_{\dot{V}O_{2\max}}$, $T_{\max}$, and the lactate threshold can be significantly improved in moderately trained runners using a 10 weeks interval training programme with intensities at $v_{\dot{V}O_{2\max}}$ and interval durations of 60% $T_{\max}$. Moreover, the present study has shown that supramaximal high-intensity interval training can significantly improve 3000 m running performance, $\dot{V}O_{2\max}$, $v_{\dot{V}O_{2\max}}$ and $T_{\max}$ in moderately trained runners. The changes in performance and physiological variables appeared more profound (in absolute, but not statistical terms) in the interval training group performing longer intervals at intensities of

$\dot{V}O_{2\max}$ with interval durations of 60% $T_{\max}$ when compared to the group performing supramaximal high-intensity interval training.

Practical applications

- Prescribing high-intensity interval training with work speeds equal to the velocity associated with maximal oxygen uptake and interval durations around 60% of the time sustainable at that velocity can improve 3000 m running performance time.
- Repeated short-term maximal sprint interval training can also improve 3000 m running performance.
- The improvement in 3000 m running performance and related physiological variables may be greater (in absolute, but not statistical terms) using more prolonged high-intensity interval training at the velocity associated with maximal oxygen uptake when compared with supramaximal sprint interval training.
- In order to reduce the monotony of a training programme, coaches may wish to use a mix of supramaximal and more prolonged interval training sessions when designing their athletes' training programmes, but superior results may be obtained using more prolonged interval durations at the velocity near maximal oxygen uptake.

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